

Scope of Work – Deployment of OEM Resources for Data Migration and Document Management System (DMS) Migration

1. Introduction

This document outlines the requirements for engaging a System Integrator (SI) to execute the Migration of the Oracle ERP database and Document Management System (DMS) of MPPMCL through the deployment of qualified Oracle OEM resources. The objective is two-fold: first, the installation, configuration, and migration of legacy database to a secure, scalable architecture (as per detailed in the scope) at the MPSEDC Bhopal Data Centre; and second, the Migration of existing NewGen DMS to seamless transfer of electronic records and metadata to MPPMCL's new centralized in-house DMS.

The Selected Bidder shall remain accountable for the security and integrity of all data throughout the Migration Lifecycle and ensure zero loss of digital assets for DMS services.

2. Project Overview

The project encompasses the deployment of Oracle OEM resources for the installation, configuration, and migration of legacy application data to new Oracle and MSSQL Server environments. The activities will be executed across both Production and UAT environments, with a focus on:

- Implementing a modern, secure, and highly available database infrastructure as per the approved architecture.
- Installing and configuring Oracle products such as RAC, APEX, ORDS, ODI.
- Migrating data from legacy Oracle E-Business Suite and custom applications to the new Oracle database(s) at MPSEDC while ensuring data integrity and accuracy.
- The Data Migration Service must involve securely transferring data from the source environment to the target system while adhering to best practices for backup, maintenance, and performance administration. The selected bidder will be responsible for resource deployment, project coordination, complete documentation and ensuring timely completion within the stipulated project duration.

The activities involve technical migration services to be delivered by the selected bidder.

- The scope also includes transitioning the existing integration/mapping between the NewGEN Document Management System (currently mapped to IABS) and IABS to a newly developed custom Document Management System, including but not limited to:
 - i The extraction and documentation of current mappings from the NewGEN Document Management System,
 - ii Transformation and implementation of equivalent mappings in the custom Document Management System, and

iii Validation of the end-to-end integration.

- The System Integrator (SI) shall be solely responsible for engaging, coordinating, and fulfilling all Original Equipment Manufacturer (OEM) requirements, dependencies, tools, and support necessary for the data migration, including but not limited to Newgen (DMS), Oracle, and Microsoft SQL Server (MS SQL)

Source Environment

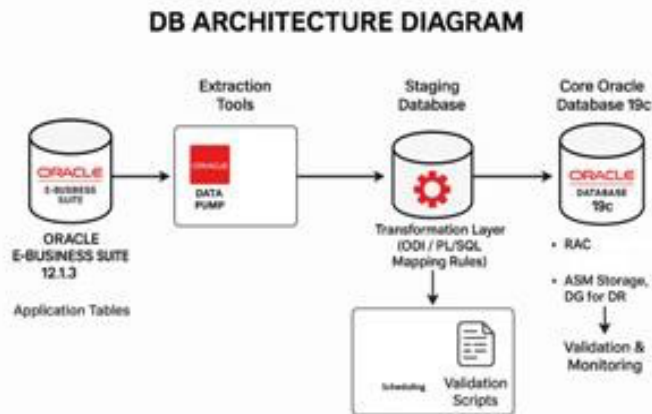
- **Application:** Oracle E-Business Suite (Finance, Procurement, HRMS, Payroll, Core, non-Core, PIMS (Power Invoice Management System)) and any module Jointly decided by MPPMCL, MPSEDC & SI.
- **Database:** Oracle 11g (11.2.0.4.0)
- **DB Size:** ~850GB (ERPPROD:500GB, FMWPROD:200GB, PTPROD:150GB)
- **OS:** Oracle Linux6.10

Target Environment

- Oracle Enterprise Edition 19c (or latest supported version) with RAC, APEX, ORDS, ODI.
- High-availability with RPO of 30 minutes and RTO of 60 minutes.
- SQL Server database instance for HRMS/Payroll (provisioned by MPSEDC).
- It will be hosted in the MPSEDC State Data Centre

3. Scope of Oracle Data Migration

The core Scope of Oracle Data Migration is to supply **Oracle OEM resources** for the installation and configuration of Oracle Enterprise Edition Database (Release 19C or latest supported) and data migration of legacy (current) application to the new application as per the requirements of the target application that is to be deployed on Oracle database and MSSQL server database. The detailed requirements are mentioned in the section below.



- The activities mentioned under the scope of work to be **carried out on the Production and UAT** environments.
- OEM resources have to implement a modern, secure, scalable, and high-availability database infrastructure and environment as per the approved Architecture that is to be hosted at Madhya

Pradesh State Electronics Development Corporation (MPSEDC) State Data Centre.

- Post installation and configurations, Oracle OEM resources need to migrate the legacy data of the Oracle E-biz application and custom application data (Running on Oracle Database) of the MP Power Management Co. Ltd., Jabalpur, to the new Oracle databases created at the MPSEDC data center. In contrast, the SQL server database instance will be provisioned (Provided) by MPSEDC.

The project involves product installation, configuration and technical migration services to be delivered by Oracle OEM resources; however, the selected bidder has to lead the accountability and ensure the resources availability, **project leadership, coordination, and successful completion within the stipulated timeline.**

If additional clarifications are required, bidder may visit the MPSEDC Bhopal

The overall objectives of this engagement include:

- Upgrade of Oracle 11G to Oracle 19C. Deployment of Oracle 19C (or latest) Enterprise Edition in high-availability and with disaster recovery setup (RPO: 30 minutes and RTO: 60 minutes) as per the approved architecture of the Project.
- Installation and configuration of Oracle products such as RAC, APEX, ORDS, ODI.
- Secure configuration of backup and monitoring systems.
- Backup plan, maintenance plan and performance administration implementation as per best practices
- Study current source and target databases' schema (and corresponding application modules) from the data migration perspective.
- Mapping source and target data structures in line with the required application (Module) framework.
- Migration of legacy Oracle databases running custom (be spoke) applications and schemas from COTS (Oracle E-Biz) to the new database environments.
- SQL server database would be provisioned and provided by MPSEDC; data migration of the specific module is required to be done.
- MPPMCL will confirm the final scope and time periods for SQL Server migration.
- Full documentation, validation, and training to be provided before handover.
- Prepare and handover Check checklist to manage Oracle environments in production after handover, including cluster and performance management.

DMS: Document Management System:

As the Document Management System (DMS) is a non-Oracle-based solution, hence System Integrator (SI) has to take a lead role in executing the DMS migration. SI may coordinate with appropriate third-party service providers to ensure a seamless transition.

The bidder is requested to share the **detailed scope for the DMS migration** at the earliest. The scope details should include, but not be limited to, the following:

- The **modules or functional areas** within the existing system that contain document repositories.
- The **total number of documents** planned for migration.
- The **types or categories of documents**, such as transactional, master data, workflow-related, and reference documents.

This information will be used to define the **migration strategy, timelines, dependencies, and validation approach** for the DMS migration activity.

Existing DMS:

Currently, the organization is using the NewGen Document Management System for document management. Under the new solution, MPPMCL will use a custom Document Management System (DMS) developed by MPSEDC to organize, store, manage, and track electronic documents and digital files through a centralized repository, enabling efficient access, retrieval, and sharing of information.

4. Scope of Document Management System (Data Migration)

To be performed by SI:

1. Conduct an AS-IS study of NewGen DMS
2. Map metadata + fields between NewGen & new DMS
3. Analyze workflow differences & align them
4. Assess infra requirements for new DMS
5. Take complete data backups
6. Review & clean exported data
7. Transform data formats for new system
8. Implement full data import process
9. Ensure metadata + file integrity
10. Test retrieval, workflow, security, user ops
11. Train end-users
12. Provide post-migration hyper care
13. Conduct post-migration audit
14. Prepare rollback plan
15. Securely shutdown/ archive old Newgen DMS

5. Deployment of System Integrator (SI)

- The **System Integrator (SI)** shall act as the **prime contractor, project coordinator, and be accountable for the overall delivery**, encompassing both Oracle and non-Oracle components (e.g., DMS Migration).
- During AS-IS, SI resources shall be deployed at MPPMCL or **MPSEDC** wherever required. They must coordinate with the OEM to draw the requirements. A dedicated Single Point of Contact (SPOC) from the SI shall be nominated and available throughout the engagement and shall be present onsite.
- The entire project execution activities, including the data platform creation and data migration, must be completed by SI under the supervision of the respective OEMs (Oracle and NewGen).
- For any support required to the system post migration till the Go-Live, the SI resources may connect remotely using any secured access (Like VPN) under written consent from MPSEDC.
- The project needs to be coordinated and accountable by the SI, and it is the bidder's responsibility to communicate and deploy the OEM resources as per the project requirements and timely delivery.

- The SI shall **ensure rigorous validation of all migration activities** and the **proper handover of all implementation and migration documentation**.

6. Deployment of OEM Resource

- The entire project execution activities, including the data platform creation and data migration, must be completed by Oracle OEM resources only.
- **On-site presence of OEM resources is mandatory during key project stages**, particularly the **initial setup/planning and final closing/Go-Live phases**.
- For the **remaining project period**, activities may be performed **remotely via secure access methods (e.g., VPN)**, subject to **mutual consent / prior agreement with the MPSEDC**.
- The SI is responsible for coordinating the project, ensuring accountability, and deploying OEM resources as required, while maintaining timely delivery and clear communication.
Ensure validation of migration and handing over of the implementation and migration documents

Note: All activities must be performed by OEM-certified resources only. The SI retains full accountability for the project's successful and timely completion.

Reporting:

- SI needs to share a weekly progress report in the format approved/provided by the MPSEDC.

7. Scope of Execution Activities.

Overall activities to be executed by the Oracle OEM resources can be considered in two critical

Parts: -

Part A: Installation and Configuration **Scope of Work**

Part B: Data Migration

The activities are detailed in below sections

Part A: Upgrade, Installation and Configuration of Oracle database

Upgrade, Installation and Configuration of Database and Cluster

- Share prerequisites to initiate Oracle configuration and installation
- Verify pre-installation checks; ensure OS prerequisites, user and group creation, and package installations (if any).
- Install Oracle 19C (or the latest available and supported database release, as agreed with department) Enterprise Edition at provided server environments.
- Listener and net services configurations
- Setting up instance parameters and profiles.
- Configuration of memory, I/O, optimizer parameters.
- Install Oracle Grid Infrastructure.
- Identify the shared storages (ASM or certified AS/SAN) and voting disks.
- Setup cluster interconnects and virtual IP addresses.
- RAC-aware database creation and Functional RAC cluster across two nodes.
- Deploy latest patches (RU/PSU), as needed.
- Enabling necessary services (flashback, archive log).

Installation of Oracle APEX and ORDS

- Deploy Oracle APEX and ORDS on a separate compute node for low-code web application development.
- Install **Oracle** APEX within the target Oracle database
- Deploy ORDS and connect with the database in standalone or web server mode.
- Create a sample workspace for validation and testing purposes.

Installation of Oracle Data Integrator (ODI)

- Enable data integration and ETL operations by deploying ODI.
- Install Studio, agents, Configure master and work repositories, as per data migration and /Or replication requirements finalized.
- **Establish** connectivity between source (legacy Oracle) and target Oracle and/or SQL Server environments, as per data migration and/or replication requirements finalized.
- Functional ODI development and execution environment.
- Test mappings and logging mechanisms.

Backup Configuration

- Develop backup scripts for full/incremental, archive, and control file and configure automated scheduling.
- Setup validation and retention policies.
- Integration with third-party backup solutions (if any).

A.6 Validation and User Acceptance Test (UAT) of the Database upgrade

Pre-Validation Steps (Before Upgrade)

- Take full backups of all databases prior to upgrade initiation
- **Pre-Upgrade Checks:** Run the Oracle pre-upgrade information tool to identify any potential issues that need addressing before the upgrade.

Post-Upgrade Validation Steps

- Check Oracle Version: Verify that the database version is indeed 19c (or latest /agreed version).
- Database Logs: Review the alert logs and trace files for any errors or warnings that occurred during the upgrade process.
- Data Integrity: Run integrity checks to ensure no data has been corrupted during the upgrade.
- Compile Invalid Objects: Compile invalid PL/SQL objects and ensure there are no invalid objects
- Check Compatibility Level: Ensure the database's compatibility level is set to 19c or Latest version.
- Performance Monitoring: Compare performance metrics before and after the upgrade to ensure there are no regressions.
- Testing Application Functionality: Conduct regression testing on applications to ensure they work correctly with 19c.

Documentation of Database Configuration Activities

Deliverables

- Maintain and document steps and procedure of configuration and installation that can be used for future configuration by department.
- Document technical parameters, backup procedures, schema designs.

Training and Knowledge Transfer (KT) Activities

- Training sessions to provide KT on the installation of the configured environment.
- Training of **designated MPSEDC/MPPMCL resources** on system management and administration
- **Provide all relevant documentation** as part of knowledge transfer.

Part B: Data Migration

The second core area involves the migration of data from legacy Oracle 11.2.0.4 environments running at Jabalpur to the newly established systems in Bhopal's SDC. This includes both COTS E-Biz Suite and custom-built applications.

<u>Activity</u>	<u>System Integrator (SI)</u>	<u>MPSEDC</u>	<u>MPPMCL</u>
Infrastructure readiness	C	R	I
Installation & configuration	R	A	C
Data migration execution	R	A	C
Testing	R	A	C
Documentation	R	A	C
Training & handover	R	A	C
Project reporting	I	R	C
Validation	R	A	C
Data Identification	R	C	A

R-Responsible, A–Accountable, C–Consulted, I–Informed

Overview of Source Systems

Database	DB Version	Appx. DB	OS
		Size (GB)	
ERPPROD-Ebiz	11.2.0.4.0	500	Oracle Linux Server release 6.10
FMWPROD-Ebiz	11.2.0.4.0	200	Oracle Linux Server release 6.10
PTPROD-Custom Java Application	11.2.0.4.0	150	Oracle Linux Server release 6.10

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Functional Areas for Data Migration

- Oracle EBS Modules: (Finance, Procurement, HRMS, Payroll).SI has to make ensure no any module shell be left out.
- Include details of **Custom Applications: PIMS, Core, and Non-Core**, covering schema structure, data mapping, and integration alignment with the target application.

OEM Data Migration Responsibilities

Execute complete migration activity from the Oracle EBS application data to the Oracle 19c database environment.

Oracle database

Understanding Source Structure

- Understand both **Source** and **Target** schema structures, including data models, relationships, and constraints, to ensure accurate data mapping and migration alignment with the new application
- Understand target schema structure aligned to new application.
- Understand the target schema design with the application development team on target schema design.
- Understand and create Document field (data structure) mapping with data types and constraints of the source and target databases. (in coordination with application development team).
- Establish referential integrity and indexes, if any.

Understanding Target Structure

- Understand target the schema structure aligned to new application.
- Understand the target schema design with the application development team on target schema design.
- Understand and create Document field (data structure) mapping, including data types and constraints between the source and target databases. (in coordination with the application development team).
- Establish referential integrity and indexes, if any.
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Data Migration

- Migrate **approval workflow data** (e.g., PIMS using ERP workflows) to ensure process continuity in the target application.
- Develop and mapping sheets detailing: Source table→ Target table/field.
- Configure data integration methodology and scripts for data migration, considering the most appropriate tools.
- Perform full and delta exports using (as per the defined migration approach taken).
- Implement data transformation logic as required by the target schema design.

- Validation of data migration.
- Create mapping sheets and document the transformation logic.
- Automation of the script for future iterations of data migration.

Note: MPPMCL will provide necessary technology licenses for database and data integration tools (ODI). All additional data elements confirmed by MPPMCL must be included in migration scope.

B.1.2 Migration to Microsoft SQL Server

- Identify cross-platform data to be moved from the Oracle EBS module to SQL Server database (tentatively the data of the Human Resource and Payroll module).
- Convert data types to match the heterogeneous platform.
- Load data using suitable tool like data integration tool or file loaders, as agreed and suitable methodology. If any other steps are defined other than the above-defined step are used, then prior approval is needed.
- Validated SQL Server instance.
- Source-to-target data mapping for SQL Server migration.
- Execute Data migration scripts activities
- **Perform comprehensive validation** of the SQL Server migration

B.1.3 Schema-Based Migration

- Migrate from Oracle database 11GR2 to latest deployment ensuring compatibility between the source and target environments.
- Perform dependency analysis for interlinked schemas.
- **Utilize appropriate tools** (ODI/Data Pump) for schema migration, as per the requirement and methodology finalized.
- **Conduct thorough data validation,**

Migration Phase and Execution

Phase 1: Initial migration

- This activity will be done for migration testing and acceptance purposes.
- Dry run of the complete migration process and scripts.
- Execute the migration scripts at all the target databases/schemas.
- Validate volume, field mapping, and target integrity.
 - Record count matching between source and target.
 - Spot-check of transactional data with functional users.
- Feedback on corporate and troubleshooting for the final run.
- Define test cases for functional validation, data validation, and user sign-off.

Note: Based on the actual scenario and test cases pre-approved by MPPMCL, MPSEDC will define and recommend the appropriate validation process to the SI.

B.2.2 Phase2: Final for Go-live

- The application downtime for final migration should not exceed 2 days (48hours), preferably to be performed on the weekend.
- Freeze source for a short window.
- Perform migration (or a complete migration)
 - Run the complete migration process and scripts.
 - Execute the migration scripts at all the target databases/schemas.

Go-live readiness validation.

- Record count matching between source and target.
- Spot-check of transactional data with functional users.

This activity may further be broken into two phases based on OEMs strategy (base data migration and subsequently delta data migration), bidder need to consider two level migrations in this phase prior to go-live.

Note: After formal communication on the approval from MPSEDC and MPPMCL. The elaborated approach may have some changes based on bidders approach and the deployment and implementation scenario then. The bidder need to adopt the suitable and appropriate data migration approach for the current scenario.

B.3 Project Acceptance and closing

Possible methodology for Validation

- Migration-Count validation
- Migration-Field-by-field and value sample checks
- Application testing on new data (by department team)
- Identify and validate external interfaces (e.g. APIs, flat file, DB link) to ensure connectivity post-migration.
- Sample worksheet development on Apex
 - Knowledge Transfer with detailed documentation for
 - Implementation and installation steps
 - Configuration parameters.
 - Migration details–source &Target
 - Migration steps.
 - Confirmation of knowledge transfer to be obtained from the aligned resources.

Note: Final acceptance of migration is contingent upon successful data validation and migration audit approval from MPPMCL.

T is the contract signing date.

<u>Sr. No.</u>	<u>Milestone</u>	<u>Description</u>	<u>Deliverables</u>	<u>Proposed Timeline</u>
<u>1</u>	Project Kick-off	Formal project initiation	Signed Project Charter, Detailed Project Plan	<u>T +7 days</u>
<u>2</u>	Environment Setup	Completion of installation & Configuration in UAT	UAT Environment, Installation Docs	T+45 days
<u>3</u>	Test Migration	Successful first data migration to UAT	Migration Test Report, Validation Logs	T+90 days
<u>4</u>	Final Migration and Production Readiness	Sign-off for go-Live	Production Environment, DR Drill Report	T+150days
<u>5</u>	Successful Go-Live	Final migration and application cutover	Go-Live Sign-off from MPPMCL	T+170days
<u>6</u>	Project Closure	Completion of KT and documentation handover	Final Project Report, All Documentation	T+180 days
<u>7</u>	Post Migration Support	3 months of post-migration support	Support on Data Migration and DMS.	T+270 days

The System Integrator (SI) shall be responsible for providing the following deliverables

- Detailed Project Plan (with tasks, dependencies and resources)
- Technical Design Document (for the target state architecture)
- Installation and Configuration Documentation
- Data Migration Strategy and Mapping Document
- Backup, Recovery, and DR Run books
- Weekly Status Reports
- Test Migration and Validation Reports
- As-Built Documentation (final state of the system)
- Knowledge Transfer Session Materials and Attendance Records